Comparison of Three Greedy Positive-Cone Reduction Algorithms on the Lambda and Physics Datasets

Experimental summary

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Abstract

This note summarises a controlled numerical comparison of three greedy algorithms that build a reduced *positive cone* $K = \operatorname{span}_+\{w_1, \dots, w_R\}$ approximating a set of nonnegative snapshots: the Cone-Projected Greedy (CPG), the Angular-Defect Greedy (ADG), and the modified Cone-Projected Greedy (mCPG). All three are evaluated on two datasets — a low-dimensional *lambda* dataset and a high-dimensional *physics* contact dataset — under identical snapshots, identical train/test conventions, and a common per-snapshot relative-error metric. We also report a dedicated study of the effect of *per-snapshot normalisation* on ADG, which is decisive on the physics dataset because of its very large spread of snapshot magnitudes.

1 Setup

Algorithms. All three methods grow a positive cone one (or, for ADG, one batch of) generator(s) at a time and project each snapshot x onto the current cone via a nonnegative least-squares (NNLS) problem $\min_{c \geq 0} \|Ac - x\|_2$.

- CPG selects, at each step, the snapshot with the largest *absolute* projection residual $\|x - \Pi_K(x)\|_2$ and adds that raw snapshot as a generator.
- ADG selects the snapshot with the largest *angle* $\theta_K(x) = \arcsin\big(\|x - \Pi_K(x)\|_2 / \|x\|_2\big)$ to the current cone. The angle is a *per-snapshot* (scale-invariant) novelty measure, so ADG's selection is intrinsically insensitive to snapshot magnitude. The two initial generators are the pair of snapshots with the largest mutual angle.
- mCPG selects like CPG (largest residual) but stores a *shifted, re-normalised* generator $\nu = (x - \Pi_K(x))/\|x - \Pi_K(x)\|_2$ instead of the raw snapshot, so each generator carries only the direction that is new relative to the current cone.

Metric. For every method we report the *per-snapshot relative residual*

$$\rho(x) \;=\; \frac{\|x - \Pi_K(x)\|_2}{\|x\|_2}, \qquad \rho_{\max} = \max_{x \in \mathcal{S}} \rho(x), \quad \bar{\rho} = \operatorname{mean}_{x \in \mathcal{S}} \rho(x), \tag{1}$$

evaluated by projecting the *original, physical-scale* snapshots onto each method's final basis. Because the positive cone is invariant to positive per-generator rescaling, the stored basis vectors are always on the original scale, making this an apples-to-apples comparison regardless of how the basis was built. We also report the basis condition number $\kappa_2 = \sigma_{\max}/\sigma_{\min}$ as a measure of numerical robustness of the reduced representation, and the cumulative fit time.

Datasets. Table 1 lists the two datasets and, crucially, the spread of snapshot magnitudes $\|x\|_2$, which turns out to govern how much the choice of selection criterion and of normalisation matters.

Table 1: Datasets used in the comparison. The magnitude ratio $\max_x \|x\|_2 / \min_x \|x\|_2$ (over nonzero snapshots) is the single most predictive quantity for the qualitative differences observed below.

Dataset	Snapshots $\times$ dim	Train/Test	$\min \|x\|_2$	$\max \|x\|_2$	Magnitude ratio
Lambda	50×57	31/19 (unseen params)	0.283	0.319	1.13
Physics	96×7676	whole dataset	9.04×10^6	5.46×10^9	603.6

For the lambda dataset the training set follows the paper grid (parameters $\mathrm{rad} - 0.65 \in \{0.15 + 0.01i \mid 0 \le i \le 30\}$), and relative residuals are reported on the 19 unseen test parameters. For the physics contact dataset the whole set of 96 snapshots (imposed displacements 0.18–1.01 mm) is used for both construction and evaluation.

2 Lambda dataset

Here all snapshot norms are within 13% of each other (Table 1), so the choice of selection criterion has little leverage. Table 2 shows the sweep at representative cone sizes R.

Table 2: Lambda dataset: unseen-test relative residual, basis condition number, and cumulative fit time versus cone size R. CPG and ADG select nearly the same snapshots and are essentially indistinguishable in accuracy and conditioning; mCPG reaches markedly lower error per component and keeps the basis far better conditioned.

R	Method	$\rho_{\max}$ (test)	$\bar{\rho}$ (test)	κ_2	Fit [s]
5	CPG	7.98×10^{-2}	4.54×10^{-2}	8.19×10^1	1.61
	ADG	7.98×10^{-2}	4.54×10^{-2}	8.19×10^1	0.13
	mCPG	7.82×10^{-2}	4.07×10^{-2}	9.04	0.11
10	CPG	7.10×10^{-2}	4.32×10^{-2}	1.53×10^3	1.80
	ADG	7.10×10^{-2}	4.32×10^{-2}	1.53×10^3	0.35
	mCPG	6.60×10^{-2}	3.66×10^{-2}	7.92×10^1	0.31
20	CPG	6.75×10^{-2}	4.19×10^{-2}	2.82×10^{10}	2.22
	ADG	6.75×10^{-2}	4.19×10^{-2}	2.82×10^{10}	0.93
	mCPG	1.93×10^{-2}	1.01×10^{-2}	2.52×10^{10}	0.75
30	CPG	6.75×10^{-2}	4.19×10^{-2}	4.78×10^{11}	2.77
	ADG	6.75×10^{-2}	4.19×10^{-2}	4.78×10^{11}	1.60
	mCPG	6.55×10^{-3}	1.41×10^{-3}	7.11×10^{10}	1.37

Observations.

- CPG and ADG produce identical relative-residual and condition-number curves across the sweep: with near-uniform norms, largest-residual and largest-angle selection pick the same snapshots. ADG is, however, consistently 3–5× faster in wall-clock fit time (batch selection and cheaper closed-form angle metrics).

- mCPG is the clear accuracy winner: at $R = 30$ its worst-case test error is 6.6×10^{-3}, roughly one order of magnitude below CPG/ADG (6.7×10^{-2}), and its mean error is $\sim 30\times$ smaller.
- mCPG also yields much better-conditioned bases (e.g. $\kappa_2 \approx 7.9 \times 10^1$ vs 1.5×10^3 at $R = 10$): its shifted, orthogonality-promoting generators avoid the near-duplicate directions that make the raw-snapshot bases of CPG/ADG increasingly singular as R grows.

3 Physics dataset

The physics snapshots span more than two orders of magnitude in norm (ratio ≈ 604), because low-displacement snapshots carry almost no contact force while high-displacement ones are very large. This is where the *selection criterion* becomes decisive. Table 3 reports the fixed-R sweep (all norms measured by the per-snapshot relative residual on the whole dataset).

Table 3: Physics dataset (whole set): per-snapshot relative residual and basis condition number versus cone size R. ADG's scale-invariant angle selection dramatically outperforms the residual-based selection of CPG and mCPG: at every R, largest-residual selection keeps chasing the few high-magnitude snapshots and leaves the small-magnitude ones badly resolved.

R	Method	$\rho_{\max}$	$\bar{\rho}$	κ_2
5	CPG	8.43×10^{-1}	9.32×10^{-2}	1.02×10^2
	ADG	1.66×10^{-1}	8.60×10^{-2}	1.30×10^3
	mCPG	8.43×10^{-1}	9.32×10^{-2}	9.58×10^1
10	CPG	8.14×10^{-1}	5.52×10^{-2}	2.91×10^2
	ADG	8.50×10^{-2}	4.30×10^{-2}	1.46×10^3
	mCPG	8.06×10^{-1}	5.28×10^{-2}	2.31×10^2
20	CPG	6.93×10^{-1}	1.98×10^{-2}	9.15×10^2
	ADG	3.02×10^{-2}	9.76×10^{-3}	2.08×10^3
	mCPG	6.85×10^{-1}	1.93×10^{-2}	2.99×10^2
30	CPG	5.20×10^{-1}	9.50×10^{-3}	2.53×10^3
	ADG	1.87×10^{-2}	4.04×10^{-3}	2.30×10^3
	mCPG	5.20×10^{-1}	9.36×10^{-3}	3.67×10^3

Observations.

- ADG is vastly superior on worst-case relative error. At $R = 30$ its $\rho_{\max} = 1.9 \times 10^{-2}$, versus 5.2×10^{-1} for both CPG and mCPG — a factor of nearly 28. ADG reaches $\rho_{\max} < 0.05$ by $R \approx 20$, a level CPG/mCPG never attain within 30 components.
- The reason is the interaction between magnitude spread and selection rule. CPG and mCPG rank candidates by *absolute* residual, so they overwhelmingly pick large-displacement snapshots and drive the mean error down while leaving the small-magnitude snapshots with large *relative* error — hence the stagnant $\rho_{\max} \approx 0.5$–0.8. ADG's angle criterion is scale-invariant and therefore attends to poorly-resolved small snapshots as readily as large ones.

- Note that MCPG's residual-based *selection* tracks CPG almost exactly on $\rho_{\max}$ here; its generator shifting improves conditioning on the lambda data but does not repair the selection-induced blind spot to small-magnitude snapshots on the physics data.
- Mean error $\bar{\rho}$ is similar across methods — all three drive the aggregate error down — which is precisely why $\rho_{\max}$ (not the mean) is the metric that exposes the difference.

4 Effect of per-snapshot normalisation on ADG

The results above concern ADG's *selection.* Its *stopping* criterion, in contrast, still compares the absolute residual against a single global scale $\varepsilon \cdot \max_x \|x\|_2$. On a wide-spread dataset this lets a small-magnitude snapshot exit the "unresolved" set — and stop influencing the algorithm — while still being 50%+ wrong in relative terms. Per-snapshot normalisation (dividing each snapshot by its own norm before the stopping/selection logic, while still storing physical-scale generators) converts this into a genuine per-snapshot relative-error stopping criterion. It is exposed as the `normalize_snapshots` flag on ADG and is off by default (no change for existing callers).

Because ADG's angle selection is provably invariant to positive per-snapshot rescaling, normalisation *cannot* change the order in which generators are chosen; it changes only when the algorithm decides a snapshot is "resolved". Its entire effect is therefore visible in the *stopping* behaviour. Table 4 sweeps the relative tolerance ε and reports, for each variant, the cone size R at which growth halts and the worst-case relative error $\rho_{\max}$ actually achieved on the whole dataset.

Table 4: Physics dataset: cone size R at which ADG halts, and the achieved worst-case relative residual $\rho_{\max}$, as a function of the relative tolerance ε, with the stopping check fit on raw versus per-snapshot–normalised snapshots. The raw variant's global-scale stopping test declares the small-magnitude snapshots "resolved" almost immediately, so it halts at $R = 2$ with $\rho_{\max} \approx 0.53$ for every $\varepsilon \geq 0.10$; only an extremely tight ε forces further growth, and even at $\varepsilon = 0.01$ ($R = 11$) it is stuck at $\rho_{\max} = 0.35$. The normalised variant keeps enriching until the per-snapshot relative error is genuinely below ε, driving $\rho_{\max}$ to its intrinsic floor ≈ 0.143.

ε	ADG raw		ADG normalised	
	R	$\rho_{\max}$	R	$\rho_{\max}$
0.30	2	0.533	4	0.294
0.20	2	0.533	5	0.166
0.15	2	0.533	7	0.143
0.10	2	0.533	8	0.143
0.05	3	0.483	13	0.143
0.01	11	0.353	37	0.143

A complementary view comes from running the physics pipeline with a single explicit relative target $\varepsilon = 0.15$. With normalisation, ADG correctly keeps enriching to $R = 7$ and reaches $\rho_{\max} = 0.143$, whereas raw ADG halts at $R = 2$ ($\rho_{\max} = 0.533$) and CPG and MCPG — whose global-scale stopping test is satisfied after a *single* vector — halt at $R = 1$ with $\rho_{\max} = 0.878$ (Table 5). This is the same magnitude-spread mechanism seen in the selection experiment, now acting on the stopping rule.

Table 5: Physics dataset, relative target $\varepsilon = 0.15$: final cone size and worst-case relative error. ADG with per-snapshot normalisation does not stop prematurely; every global-scale variant does.

Method	R	$\rho_{\max}$
ADG (normalised)	7	1.43×10^{-1}
ADG (raw)	2	5.33×10^{-1}
CPG	1	8.78×10^{-1}
mCPG	1	8.78×10^{-1}

Caveat. Normalisation is *data-dependent*, not a universal improvement. On the lambda dataset (magnitude ratio 1.13) the raw and normalised variants are identical at almost every tolerance, and normalisation occasionally costs one extra component and a slightly worse condition number. It is therefore kept as an opt-in flag rather than the default.

5 Conclusions

- **When snapshot magnitudes are uniform (lambda):** CPG and ADG are essentially equivalent in accuracy (ADG faster to fit); mCPG is the best method, delivering roughly an order-of-magnitude lower worst-case error and substantially better-conditioned bases thanks to its shifted, orthogonality-promoting generators.

- **When snapshot magnitudes span orders of magnitude (physics):** ADG is decisively best on worst-case relative error, because its angle selection is scale-invariant and does not neglect small-magnitude snapshots the way the residual-based selection of CPG/mCPG does ($\rho_{\max} \approx 0.02$ for ADG vs ≈ 0.52 for the others at $R = 30$).

- **Normalisation** closes the analogous gap in ADG's stopping rule: without it, the global-scale stopping test halts the algorithm at $R = 2$ with $\rho_{\max} \approx 0.53$ on the physics data (small snapshots wrongly declared resolved), whereas per-snapshot normalisation keeps enriching to the intrinsic accuracy floor $\rho_{\max} \approx 0.143$. It is effectively a no-op on the well-scaled lambda data.

- **Overall guidance:** choose mCPG for well-scaled data where conditioning and error-per-component matter; choose ADG (with `normalize_snapshots=True`) for data with a wide spread of snapshot magnitudes where uniform *relative* accuracy across all snapshots is the goal.

Reproducibility. Lambda results: `results/lambda/component_sweep/`. Physics baseline (fixed-R sweep): `results/physics/reduction/`. Physics ADG normalisation study (Tables 4 and 5): `results/physics/adg_normalization/` (`physics_adg_normalization_stop_points.csv`, `physics_eps015_meth` All numbers in this document are taken verbatim from the `*_metrics.csv` and report files in those directories.